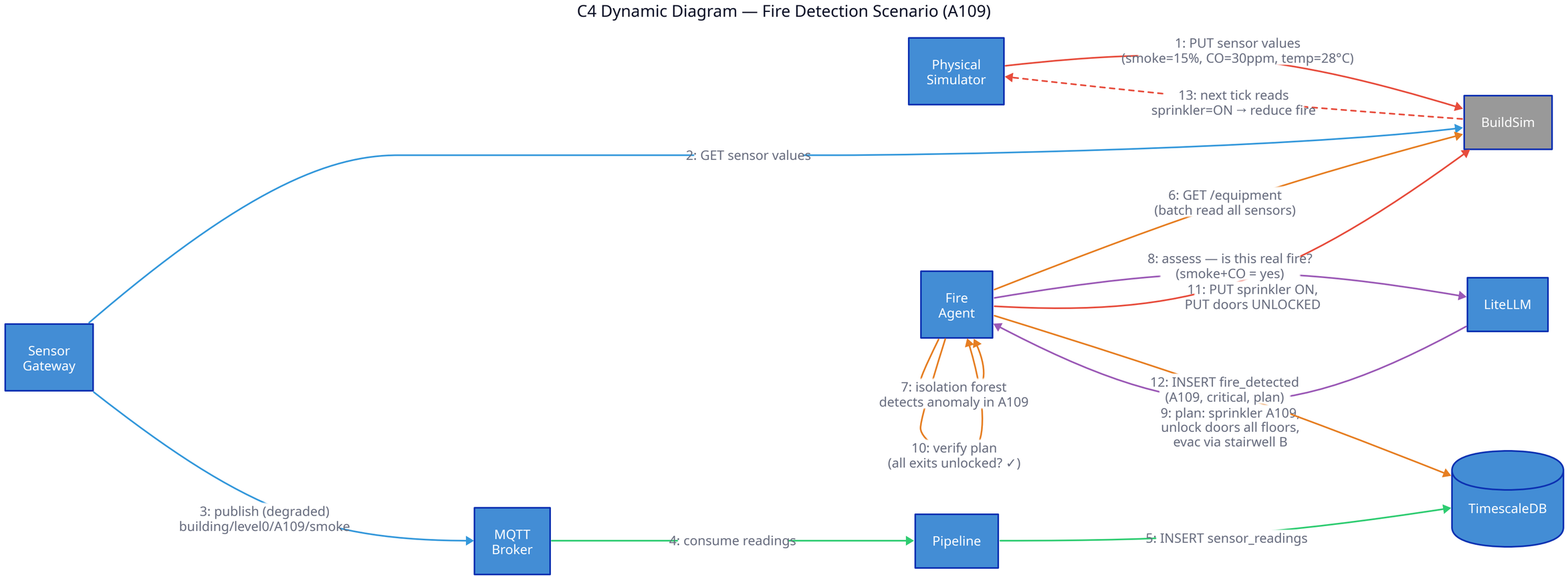

C4 Dynamic Diagram — Fire Detection Scenario (A109)
Physical Simulator
1: PUT sensor values
(smoke=15%, CO=30ppm, temp=28°C)
13: next tick reads
sprinkler=ON → reduce fire
BuildSim
2: GET sensor values
6: GET /equipment
(batch read all sensors)
8: assess — is this real fire?
(smoke+CO = yes)
11: PUT sprinkler ON,
PUT doors UNLOCKED
Fire Agent
LiteLLM
Sensor Gateway
7: isolation forest
detects anomaly in A109
10: verify plan
(all exits unlocked? ✓)
12: INSERT fire_detected
(A109, critical, plan)
9: plan: sprinkler A109,
unlock doors all floors,
evac via stairwell B
TimescaleDB
3: publish (degraded)
building/level0/A109/smoke
MQTT Broker
4: consume readings
Pipeline
5: INSERT sensor_readings